FINAL EXAMINATION
SEMESTER III, SESSION 2020/2021

COURSE CODE	:	FSPP 0034
COURSE	:	PHYSICS 1I
PROGRAMME	:	FOUNDATION UTM
DURATION	:	3 HOURS
DATE	:	11 APRIL 2021

INSTRUCTION TO CANDIDATES:

1. ANSWER ALL QUESTIONS.

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This examination paper consists of (12) printed pages only including this page

PART B

QUESTION 4 (20 MARKS)

a) In Young's double slit experiment, the slit separation is 0.25 mm and the screen is 0.15 m from the slits. When the slits are illuminated by a light, the third dark fringes are observed 9.5 mm from the central bright fringe. Calculate:

   i. The wavelength of the light. 6.33 x 10 ^-6 m

   ii. The separation between two consecutive fringes. 3.8 x 10^-3 m

   iii. How far the screen should be moved from its original position if the fringe separation is changed to 5.50 mm and the other parameters are maintained. 0.07 m

(6 marks)

b) Light from sodium lamp of wavelength 580 nm is shined directly on a narrow slit with a width of 0.02 mm. Determine the angular width of the central bright fringe. 3.32

$\theta_w = 2\theta_1 = 2\left(\sin^{-1}\frac{\lambda}{a}\right)$ $w = 2y_1 = \frac{2\lambda L}{a}$

(2 marks)

c) The fourth order line for 650 nm light falling on the diffraction grating is observed at 15.5° angle. Determine:

   i. The number of lines per cm for the grating. 1027.8 line/cm $d = \frac{1}{N}$

   ii. The angle of the second order line. 7.7

(5 marks)

d) When an electromagnetic radiation of wavelength 500 nm is incident on a metal surface, photoelectrons are emitted. If the photoelectrons are stopped by a voltage of 2.0 V, calculate:

   i. The work function of this metal in eV. 0.47 eV

   ii. The de Broglie wavelength of the electrons. $\boxed{8.68 \text{ x } 10 \text{ ^-10m}}$ 8.68×10^{-10} m

(7 marks)

4) (a) (i) $x_m = (m + \frac{1}{2})\frac{\lambda L}{d}$

$9.5\times10^{-3} = (2+\frac{1}{2})\frac{\lambda(0.15)}{0.25\times10^{-3}}$

$9.5\times10^{-3} = 1500\lambda$

$\lambda = 6.33\times10^{-6}\,m$

(ii) $\Delta y = \frac{\lambda L}{d} = \frac{6.33\times10^{-6}(0.15)}{0.25\times10^{-3}}$

$= 3.8\times10^{-3}\,m$

(iii) $5.5\times10^{-3} = \frac{6.33\times10^{-6}(L)}{0.25\times10^{-3}}$

$= 0.217$

$0.217 - 0.15 = 0.067m$

(b) $\lambda = 580\times10^{-9}m$, $a = 0.02\times10^{-3}m$

$\theta = 2\sin^{-1}(\frac{\lambda}{a})$

$= 2\sin^{-1}\left(\frac{580\times10^{-9}}{0.02\times10^{-3}}\right)$

$= 3.32°$

(c) (i) $KE = PE$

$E = W_0 + K_{max}$

$W_0 = E - K_{max}$

$W_0 = E + PE$

$= \frac{hc}{\lambda} + eV$

$= \frac{6.63\times10^{-34}(3\times10^8)}{500\times10^{-9}} + 1.6\times10^{-19}(2) = 0.49\,eV$

(ii) $\frac{1}{2}mv^2 = eV$

$v = 8.38\times10^5\,m/s$

$\lambda = \frac{h}{mv} = \frac{6.63\times10^{-34}}{9.11\times10^{-31}(8.38\times10^5)}$

$= 8.68\times10^{-10}\,m$

(d) $\lambda = 650nm$, $\theta = 15.5$, $n = 4$

(i) $d\sin\theta = n\lambda$

$d = \frac{4(650\times10^{-9})}{\sin(15.5)}$

$= 9.73\times10^{-6}\,m$

$d = \frac{1}{H}$

$H = \frac{1}{9.73\times10^{-6}}$

$= 102784$ lines per meter

(ii) $d\sin\theta = n\lambda$

$9.73\times10^{-6}\sin\theta = 2(650\times10^{-9})$

$\sin\theta = 0.1336$

$\theta = 7.68°$

QUESTION 5 (20 MARKS)

a) Radioactive decay is a random and spontaneous nuclear reaction. Explain the terms random and spontaneous.

(2 marks)

b) A laboratory has 2.56 μg of pure $^{214}_{83}\mathrm{Bi}$, which has a half-life of 20 min. Determine:

i. The number of nuclei present initially. 7.2 x 10 ^15

ii. The initial activity. 4.16 x 10 ^12

iii. The activity after 1.00 hr. 5.19 x 10 ^11

iv. The time, t when $\left|\frac{dN}{dt}\right| = 1.00\ \mathrm{s}^{-1}$. 13.96h

(8 marks)

c) Equation (1) describe the nuclear fission reaction of Uranium-235 nuclei by slow moving neutrons. Calculate the energy released in the reaction. 211 MeV

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} = {}^{139}_{54}\mathrm{Xe} + {}^{95}_{38}\mathrm{Sr} + 2({}^{1}_{0}\mathrm{n}) + \text{energy} \qquad (1)$$

(2 marks)

d) An atomic bomb uses 180 kg of U-235. If all of U-235 in the atomic bomb undergo fission as stated in Equation (2),

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{236}_{92}\mathrm{U} \rightarrow {}^{141}_{55}\mathrm{Cs} + {}^{93}_{37}\mathrm{Rb} + 2({}^{1}_{0}\mathrm{n}) + Q \qquad (2)$$

Determine:

i. The mass loss in the fission reaction. 0.19405 u

ii. The total reduction of mass of U-35 in atomic bomb. 8.95 x 10^25 u

(4 marks)

e) i. Why does a fusion reaction take place at high temperature?

ii. Two deuteron fuses together to form a triton and a proton as stated in Equation (3). Calculate the energy released. 4.03 MeV

$$^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{3}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} + Q \qquad (3)$$

(4 marks)

5) (b) (i) $214g \Rightarrow 6.02\times10^{23}$ nuclei

$2.56\times10^{-6}g \Rightarrow 7.201\times10^{15}$ nuclei (N)

(ii) $T_{\frac{1}{2}} = \frac{\ln(2)}{\lambda}$ $\qquad$ $A_0 = \lambda N_0$

$\lambda = 5.78\times10^{-4}\,s^{-1}$ $\qquad$ $= 5.78\times10^{-4}\times7.201\times10^{15}$

$= 4.16\times10^{12}\ d/s$

(iii) $A = A_0 e^{-\lambda t}$

$= 4.16\times10^{12}\times e^{(5.78\times10^{-4})(60\times60)}$

$= 5.193\times10^{11}\ d/s$

(iv) $A = \left|\frac{dN}{dt}\right| = 1.0$ $\qquad$ $1730 = 7.2\times10^{15}e^{-5.78\times10^{-4}\times t}$

$5.78\times10^{-4} = \frac{\frac{dN}{dt}}{N}$ $\qquad$ $\ln(2402\times10^{-16}) = -5.78\times10^{-4}t$

$N = 1730$ $\qquad$ $-29.06 = -5.78\times10^{-4}t$

$t = 50271s = 13.964$ hour

(c) $Q = (235\times7.60) - (139\times8.39 + 95\times8.74)$

$= 1786 - 1996.51$

$= -210.51$ MeV

(d) (i) $\Delta m = (235.04392 + 1.00867) - (140.91963 + 92.92157 + 2(1.00867))$

$= 0.194054$ (mass defect for one nucleus U-235)

(ii) $0.235kg \rightarrow 6.023\times10^{23}$ nuclei

$180kg \rightarrow 4.613\times10^{26}$ nuclei

$\therefore$ mass defect for 180 kg U-235 $= 0.194054\times4.613\times10^{26} = 8.95\times10^{25}u$

(e) (i) Combine two small nuclei to form a larger nucleus

(ii) $\Delta m = m_i - m_p$

$= (2.014102\times2.014102) - (3.016049 + 1.007825)$

$= 4.33\times10^{-3}u$

$Q = \Delta mc^2$

$= (4.33\times10^{-3}\times931.5)(3\times10^8)^2$

$= 4.03$ MeV

APPENDIX 1

List of constant

$p = +1.60 \times 10^{-19}$ C

$e = -1.60 \times 10^{-19}$ C

$\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}

$\varepsilon_0 = 8.85 \times 10^{-12}$ C^{2}N^{-1}m^{-2}

$g = 9.80$ m s^{-2}

$c = 3 \times 10^8$ m s^{-1}

$m_e = 9.11 \times 10^{-31}$ kg

$k = 9.0 \times 10^9$ N m^2 C^{-2}

$h = 6.63 \times 10^{-34}$ J s

$N_A = 6.022 \times 10^{23}$

1 u = 1.6605×10^{-27} kg = 935 MeV/c^2

1 eV= 1.60×10^{-19} J

Resistivity of iron at 20 °C: 9.71×10^{-8} Ωm

Resistivity of tungsten at 20 °C: 5.6×10^{-8} Ωm

Resistivity of platinum at 20 °C: 10.6×10^{-8} Ωm

Temperature coefficient of iron, α: 0.0651 °C^{-1}

Temperature coefficient of tungsten, α: 0.045 °C^{-1}

Temperature coefficient of platinum, α: 0.00397 °C^{-1}

APPENDIX 2

List of Formula

$$E = k\frac{Q}{r^2}$$

$$F = k\frac{Q_1Q_2}{r^2}$$

$$U = k\frac{qQ}{r}$$

$$V = k\frac{Q}{r}$$

$$W = q\,V$$

$$\Delta U = q\,\Delta V$$

$$\Delta V = -Ed$$

$$F = q\,vB\sin\theta$$

$$B = \frac{\mu_0 I}{2\pi d}$$

$$F = ILB\sin\theta$$

$$U = \frac{1}{2}LI^2$$

$$P = IV$$

$$\rho_T = \rho_o[1 + \alpha(T - T_o)]$$

$$j = \frac{I}{A}$$

$$I = \frac{Q}{t}$$

$$M = \frac{\mu_o N_1 N_2 A}{I}$$

$$L = \frac{\mu_0 N^2 A}{l}$$

$$I = I_{max}\left(1 - e^{-t/\tau}\right) = \frac{V}{R}(1 - e^{-t/\tau})$$

$$Q = Q_o e^{-t/\tau}$$

$$Q = Q_o(1 - e^{-t/\tau})$$

$$I = I_0 e^{-t/\tau}$$

$$\varepsilon = -L\frac{dI}{dt}$$

$$\tau = \frac{L}{R}$$

$$\varepsilon = -\frac{d\emptyset_B}{dt}$$

$$\frac{V_S}{V_P} = \frac{N_S}{N_P}$$

$$v = f\lambda$$

$$E = Bv$$

$$k = \frac{2\pi}{\lambda}$$

$$P = \frac{1}{f}$$

$$m = -\frac{d_i}{d_0} = \frac{h_i}{h_0}$$

$$\frac{1}{f} = \frac{1}{d_0} + \frac{1}{d_i}$$

$$\frac{1}{f} = (n-1)\left[\frac{1}{r_1} + \frac{1}{r_2}\right]$$

$$n_1\sin\theta_1 = n_2\sin\theta_2$$

$$d\sin\theta = m\lambda \quad , m = 0,1,2,3\ldots$$

$$d\sin\theta = \left(m + \frac{1}{2}\right)\lambda \quad , \quad m = 0,1,2,3\ldots$$

$$Q = \Delta m\,c^2$$

$$KE = \frac{hc}{\lambda} - W_0$$

$$\left|\frac{dN}{dt}\right|_0 = \lambda N_0$$

$$\left|\frac{dN}{dt}\right| = \left|\frac{dN}{dt}\right|_0 e^{-\lambda t}$$

$$\lambda = \frac{0.693}{T_{1/2}}$$

$$\mathcal{E} = \left|-M\frac{dI}{dt}\right|$$

$$I = \frac{\varepsilon}{R}$$

$$\varepsilon = NB\omega A$$

$$\varepsilon = Blv$$

$$B_o = \frac{E_o}{c}$$

APPENDIX 3

Nuclide	Binding energy per nucleon, *E* (MeV)
$^{95}_{38}Sr$	8.74
$^{139}_{54}Xe$	8.39
$^{235}_{92}U$	7.60

Nuclide	Binding energy per nucleon, *E* (J)
$^{144}_{56}Ba$	1.3341×10^{-12}
$^{90}_{36}Kr$	1.3864×10^{-12}
$^{235}_{92}U$	1.2191×10^{-12}

Nuclide	Mass, *m* (u)
$^{235}_{92}U$	235.04392
$^{141}_{55}Cs$	140.91963
$^{93}_{37}Rb$	92.92157
$^{1}_{0}n$	1.00867
$^{1}_{1}H$	1.007825
$^{2}_{1}H$	2.014102
$^{3}_{1}H$	3.016049
$^{3}_{2}He$	3.01605